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Embedded systems

Smart

**Door Lock System Using Arduino
and RFID Technology**

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ABSTRACT

The *Smart Door Lock System* is an innovative project using Arduino and RFID technology to enhance home security. The system identifies authorized individuals through RFID tags and controls a servo motor-based locking mechanism. An LCD

provides real-time feedback on system status, and LEDs signal access approval or denial. This report describes the design, implementation, and societal impact of a Smart Door Lock System using Arduino and RFID technology. The system enhances security by employing RFID cards to grant or deny access to a door. The report includes details on the hardware components, software code, methodology, and experimental results, showcasing the effectiveness of the project in improving access control systems. Additionally, the societal impact of the project is discussed

INTRODUCTION

This project is made by **Farhad Ahmed from** BE.CSE-V Sukkur IBA University. Smart access control systems are becoming increasingly essential in modern security applications. Traditional locks often lack sufficient protection against unauthorized access. This project aims to provide a cost-effective and user-friendly solution to address these limitations by integrating Arduino and RFID technology. To start, let's connect the RFID sensor, LED/buzzer, and servo to Arduino using a couple of jumper wires. This will enable us to test the circuits and to read the serial numbers from my tags to load them into the array in my code so that it will open the lock.

List of Equipment

- Resistors 220 Ohm / 10k Ohm
- LED
- Arduino UNO
- Buzzer
- A couple of jumper Wires
- Bread Board
- RFID tag Module
- Door Lock
- Servo Motor

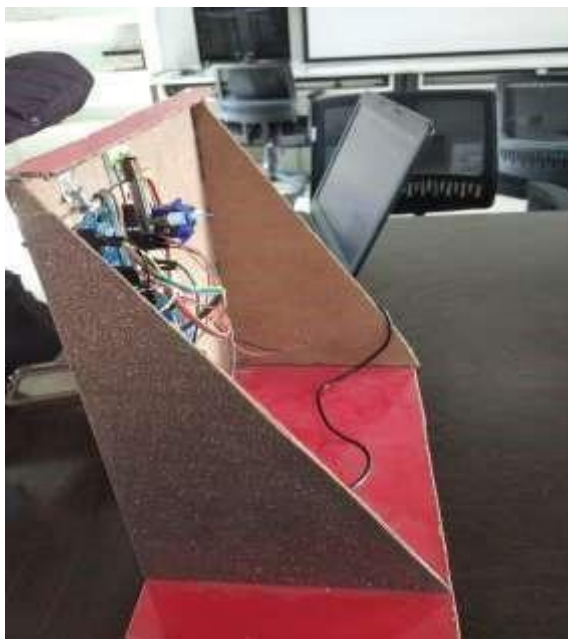
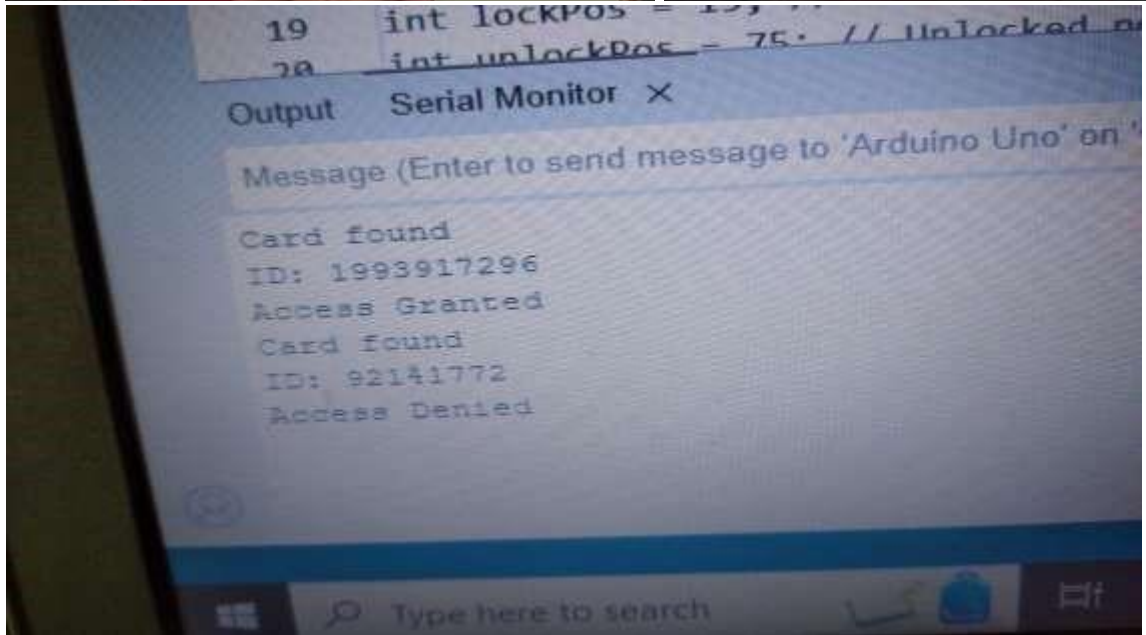
Review of Past Work

RFID technology has been widely used in various applications such as inventory management, public transportation, and security systems. Previous studies highlight the effectiveness of RFID in enhancing secure access. However, these solutions often lack affordability or adaptability. This project builds on

these insights to deliver a more efficient and scalable design.

The Circuit

The RC522 sensor is going to be connected to our Arduino using the SPI interface. We are then going to connect a green LED which will flash to show that a tag has been read and access has been granted and a red LED which will flash to indicate that a tag has been read an access has not been granted. Lastly, a micro servo will be used to open and close the locking mechanism. The LEDs and micro servo are connected using typical Arduino circuits.







Methodology

The Smart Door Lock System consists of the following main components:

1. **RFID module:** Reads the unique ID of RFID cards.
2. **Arduino Uno:** Processes the RFID input and controls the servo motor.
3. **Servo motor:** Acts as the locking mechanism.
4. **16x2 LCD:** Provides user feedback.
5. **LED indicators:** Indicate access status.
6. **Buzzer:** Alerts users in case of unauthorized access attempts.

Workflow

- The RFID reader scans the presented card.
- The Arduino matches the scanned card ID with pre-stored IDs.
- If the card is authorized, the servo motor unlocks the door and an indicator light turns green.
- If the card is unauthorized, the system denies access, flashes a red light, and sounds the buzzer.

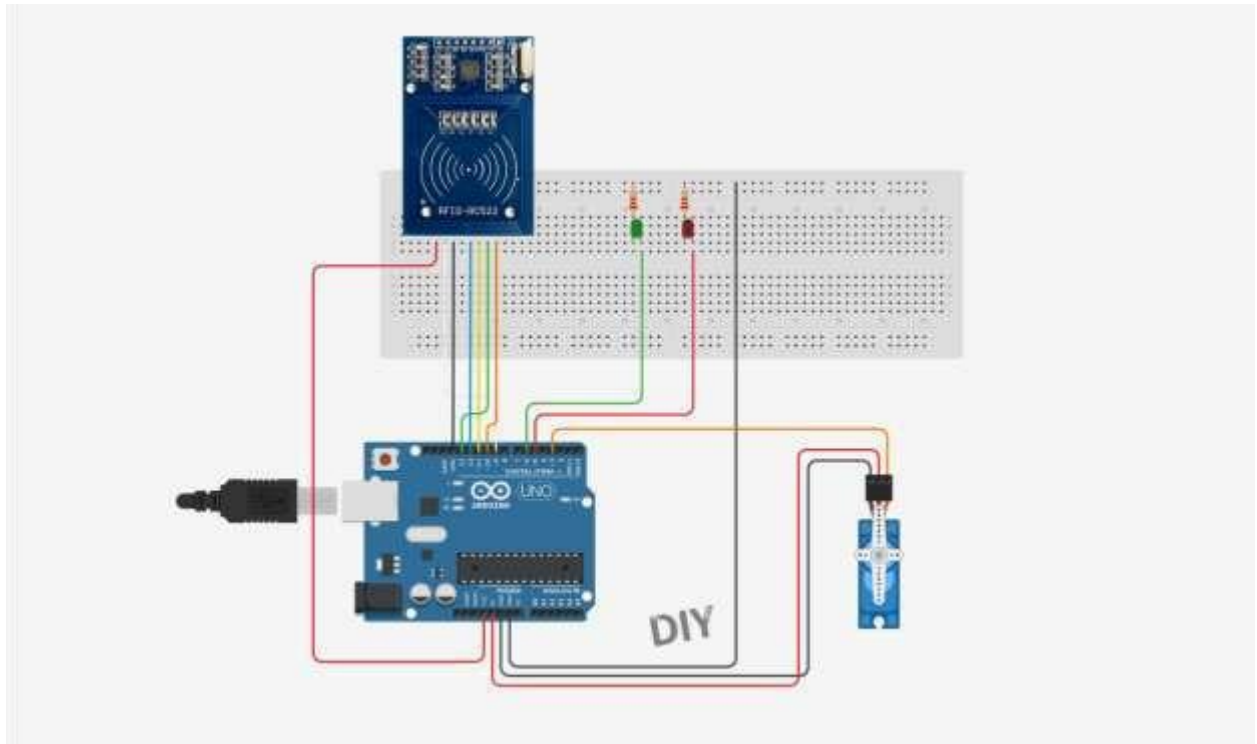
Experimental Results

The system was tested with a variety of RFID cards:

1. Authorized cards successfully unlocked the door.
2. Unauthorized cards triggered a denied access alert, with the red LED and buzzer activated.
3. The LCD displayed clear and accurate feedback for each interaction.



Circuit Diagram



Building the Lock Mechanism

Once you've got the circuit connected on your breadboard/jump wires, you'll need to print and assemble your lock mechanism.

I use a simple door lock, you can use any standard bolt type sliding lock available from your local hardware store. You'll just need to attach one end of the servo push rod to the bolt to actuate it. You'll also need to build a simple servo bracket to hold the servo in place behind the lock. You can just glue the servo in place, but a screwed on bracket is usually a bit stronger. The picture of my door lock which I have used in this project are as under.



Uploading The Code

Once you've assembled your lock mechanism, you're ready to upload your code and load your tag numbers into the array.

NOTE. Before you upload your code, you'll need to install the RFID library which is bundled with the sketch zip file. This is

easily done in your Arduino IDE by clicking on Sketch -> Include Library -> Add .ZIP Library and then selecting the zipped library file.

In the code we first include the required libraries and then set up the sensor object and an array to return the read tag serial number. The next array and its associated size is used to store the serial numbers for all of the tags which you'd like to grant access to. You'll need to find and update these numbers using the serial monitor by uploading this code and then scanning your tags. The Serial monitor will display the tag's serial number and then state that access is denied. Copy this number into the array `accessGranted`, update the array size (number of tags registered) and then re-upload the code. You could also write a short section to enable you to register a new tag by pushing a button inside the sensor component box or inside the door for example. We then set up the servo object and its travel limits. You may need to make adjustments to these

limits to get your servo to move through its full range without overtravelling in either direction. In the setup code, we connect to the RFID sensor, define the LED pins and then run through a quick LED flash startup sequence before making sure that the lock is in the locked position. You can remove the Serial monitor output lines in the code in your final version, these are just useful for registering your tags and debugging the system when you first assemble it. We then run through the loop which waits for a card or tag to be scanned, determines its serial number and then passes this serial number through to a function called `check Access` to verify whether the tag number can grant access or not. The `check Access` function simply takes the read tag number and then cycles through the array of accepted numbers to see if it is an accepted tag. If a match is found then the green LED is flashed and the lock is either opened or closed, depending on the

previous state. If the tag number is not found in the array then the red LED is flashed and the lock is not opened.

Adding or Removing Accepted Tags

As mentioned in the previous step, the array access Granted is used to store the serial numbers of the accepted tags and the integer access Granted Size stores the number of entries in this array.

To add a tag, you'll need to scan the tag with the Serial monitor open on your PC. You'll then get a message saying that the card has been read along with the serial number of the card and a message saying "Access Denied". Simply copy this number into the array and update the size integer to reflect the new number of tags. Re-upload the code and you should now get an "Access Granted" message on the Serial monitor.

To remove a tag, find the tag serial number in the array (you may need to scan it has done previously) and remove it from the array. Update the array size for any tags which you remove as well.

It is fairly easy to include a push-button or two on the inside of the door or in the component box which will allow you to add or remove tags without having to update the code.



You should now have a functioning RFID locking mechanism which can be easily installed onto a door, cupboard or container to restrict access to it. You can also put the sensor components into a simple container or housing to mount on the front side of the door.

There are a couple of ways to make this lock a bit more secure if you're actually going to be using it to secure a room or cupboard.

Start by replacing the 3D printed lock with a proper metal lock from a hardware store. Make sure that you have a solid connection between the lock and the servo and try to position the servo such that the arm is in line with the push-rod and the head of the slider when it is in the locked position. This will ensure that you can't slip a thin object through the gap in the door and try to push the slider open, you'll be pushing against the centre of the servo and not relying on the torque provided by the servo to keep the slider in place.

Next, place as few of the electronic components outside as possible. It is better to have the actual Arduino and servo connection inside the room or box and place only the RFID sensor and LEDs outside. It's much more difficult to trick the Arduino into opening the lock using the RFID sensor connection than it is to simply provide a PWM signal to the servo to unlock the door.

APPENDIX :

Header Files and Initialization

//These libraries enable RFID communication, servo motor control, and LCD interfacing.

#include <SPI.h>

#include <MFRC522.h>

#include <Servo.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

Constants and Pins

//Defines pin configurations for RFID, LEDs, and servo motor.

#define RST_PIN 9

#define SS_PIN 10

int redLEDPin = 2; int

greenLEDPin = 6;

Setup Function

//Initializes communication, pin modes, and displays the initial status. **void**

setup() {

Serial.begin(9600);

SPI.begin();

```
    rfid.PCD_Init();  
    pinMode(redLEDPin, OUTPUT);  
    pinMode(greenLEDPin, OUTPUT);  
    lcd.init();    lcd.backlight();  
    lockServo.attach(3);  
    lockServo.write(lockPos);  
    lcd.print("Place Card...");  
}
```

Main Loop

```
//Continuously scans for RFID cards and processes their IDs. void  
loop() {  
    if (rfid.PICC_IsNewCardPresent()  
    && rfid.PICC_ReadCardSerial()) {  
        String temp = "";  
        for (byte i = 0; i < rfid.uid.size; i++) {  
            temp += String(rfid.uid.uidByte[i], DEC);  
        }  
        checkAccess(temp);  
        rfid.PICC_HaltA();  
    }  
}
```

```
rfid.PCD_StopCrypto1();  
  
}  
  
}
```

Access Control

```
//Matches the scanned ID with stored IDs and executes the locking  
mechanism. void checkAccess(String temp) {    boolean granted = false;    for  
(int i = 0; i < accessGrantedSize; i++) {        if (accessGranted[i] == temp) {  
granted = true;            lockServo.write(unlockPos);            delay(3000);  
lockServo.write(lockPos);  
        }    }    if (!granted) {  
digitalWrite(redLEDPin, HIGH);    delay(200);  
digitalWrite(redLEDPin, LOW);  
    }  
}
```



Impact on Society

The Smart Door Lock System has several societal benefits:

1. **Enhanced Security:** Provides secure access to homes, offices, and restricted areas.
2. **Convenience:** Eliminates the need for traditional keys, reducing the risk of misplacement.
3. **Affordability:** Offers a cost-effective alternative to commercial smart locks.
4. **Accessibility:** Promotes the use of smart technology in everyday life, making it more widespread.

Conclusions

The project successfully demonstrates the use of RFID technology for secure access control. The system is reliable, efficient, and adaptable for various applications. Future enhancements may include adding biometric authentication or a mobile app interface.

References

1. Arduino Documentation: <https://www.arduino.cc>
2. MFRC522 Library: <https://github.com/miguelbalboa/rfid>
3. LiquidCrystal_I2C Library:
<https://bitbucket.org/fmalpartida/newliquidcrystal/wiki/Home>

4. Servo Motor Control Basics:

<https://www.servomotorbasics.com>